ESHA HOODA

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Seeking Summer 2027 Software Engineering or Machine Learning Internship

EDUCATION

B.S. Data Science, Minors in Statistics and Computer Science, Concentration in Public Health - San Jose State University Dec 2027
Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management, Statistics & Probability, Advanced Python, OOP, Linear Algebra | Certificate: Fundamental AI Algorithms and Applications in Computer Science, SJSU (expected May 2027)

SKILLS

Languages: Python, SQL, Java, C++, JavaScript, R, HTML/CSS
ML & AI: PyTorch, TensorFlow/Keras, Hugging Face Transformers, Scikit-learn, XGBoost, RAG, LLM agents & tool-calling, embeddings + vector search, LLM evaluation, Claude API
Software & Data: Flask, REST APIs, MySQL, SQLite, Streamlit, PySpark, ETL, Pandas, NumPy, Docker, GCP, UNIX, Git, Tableau, Power BI
Tools: Claude Code, GitHub Copilot, Jira, Linear, Figma, Google Analytics, Optimizely

EXPERIENCE

Deep Learning Intern, Hewlett Packard Enterprise & National University of Singapore Dec 2022 - Jan 2023
- Trained a posture-detection model to **93% accuracy** through neural network architecture tuning and iterative experimentation
- Built **6 model types** (ANN, RNN, GAN, SVM, NLP, time series) and moved models from notebooks into **Azure** cloud deployment pipelines

Outreach Lead, Peer Connections, SJSU May 2025 - Jan 2026
- Scaled a campus initiative from **0 to 3,000 participants in 2 months** by redesigning internal processes and building **SQL-based tracking systems** while coordinating a team of 120+

Technical Learning Assistant, Peer Connections, SJSU Jan 2025 - Jan 2026
- Mentored **250+ students** in math and CS and co-designed equity-focused learning activities with professors; CRLA Level 1 & 2 certified

RESEARCH EXPERIENCE

Undergraduate Machine Learning Research Assistant (Advisor: Dr. Tahir Bachar Issa), San José State University Oct 2026 - Present
- Applying **diffusion models (SR3)** to video-rate fluorescence microscopy super-resolution, replacing the GAN-based approach from a 2020 IEEE BIBE paper that serves as the project baseline

Undergraduate Machine Learning Research Assistant (Advisor: Dr. Cassandra Paul), Research Foundation, SJSU May 2025 - Present
- Evaluate the impact of SJSU's Learning Assistant Program on **70K+ student records**: logistic regression, OLS with course fixed effects, **hierarchical linear models**, and **propensity score matching** in Python; validated survey subscales with PCA and factor analysis; testing for an instructor effect; manuscript in preparation for a conference proceeding

PROJECTS

LegalBee | Tools Used: Python, RAG (embeddings + vector search), LLM prompting, NLP classification Sep 2026 - Present
- Lead and build a **jurisdiction-aware RAG copilot** (SJSU Responsible Computing Club) that answers legal questions from **3 source types** (government, court, and legal-aid) with **citations and explicit confidence signals**; first vertical is California tenant law, with **2 evaluation metrics** (retrieval accuracy, hallucination rate) built in from the start
- Act as de facto **project manager**: set the **roadmap and sprint priorities**, delegate workstreams across teammates, and run **weekly sprints and check-ins** to keep product, design, and engineering in sync

Daybeam | Tools Used: Python, LLM tool-calling, ReAct, beam search, A* route planning, Streamlit Oct 2026 - Present
- Developing an **AI planning agent** that turns a multi-constraint request ("plan my Saturday in San José: farmers market before noon, lunch under $20, home by 6") into a full day plan; tool calls to an **A* route planner**, a places lookup, and a code-based checker verify **every constraint** (time windows, budget, travel time, opening hours)
- Evaluating **3 planning strategies** (one-shot, ReAct, beam search) on **hundreds of** multi-constraint tasks for success rate, cost, and speed to answer whether extra search is worth its cost

AI Job Application Tracker Agent | Tools Used: Python, Flask, SQLite, Anthropic Claude API (tool-calling), RAG, JavaScript, HTML/CSS Aug 2026
- Built a full-stack **agent with 5 tools** that decides on its own which tools to call (parse posting, fetch resume, score fit, log result) via Claude **tool-calling**; persisted conversation memory and a scoring prompt written to avoid inflating fit scores

Trust & Safety Risk Pipeline | Tools Used: Python, SQL, PySpark, XGBoost, Hugging Face Transformers, PyTorch, Docker Jul 2026
- Designed a **multimodal risk-detection pipeline** combining **3 signal types** (tabular fraud, NLP content, image risk) into **1 unified risk score**

Stock Volatility Prediction Pipeline | Tools Used: Python, PySpark, SQL, ETL, GCP, Pandas, UNIX, Git May 2026
- Engineered a distributed **PySpark data pipeline on GCP** processing 600,000+ daily trades across 500+ stocks; random forest feature importance identified prior-day intraday range as the dominant volatility predictor (0.837)
- Served as lead engineer and **PM** across several sprints: defined the feature roadmap, drove project vision, and aligned stakeholders through weekly report-outs and dashboards

Fake News Detection Using LSTM & BERT | Tools Used: Python, TensorFlow/Keras, PyTorch, Hugging Face, Scikit-learn, CUDA May 2026
- Compared **4 NLP architectures** (SVM, Random Forest, BiLSTM, fine-tuned BERT) on **20,000 articles**; traced near-perfect scores to a source-style confound inflating every model and recommended cross-source validation; performed GPU performance tuning

CERTIFICATIONS & AWARDS

Machine Learning & AI (IIT Bombay / IIT Madras) | Code in Place (Stanford) | IRB Research Training | CRLA Levels 1 & 2 | State Rank 2 (3 consecutive years) in IEO; Top 150 nationwide, National Children's Science Congress